

FB_Inference — generated SCL excerpt (B-spline LUT)

```
01 FUNCTION_BLOCK FB_Inference
02 VAR_INPUT
03     features : ARRAY[0..27] OF REAL;
04 END_VAR
05 VAR_OUTPUT
06     class_id : INT; confidence : REAL;
07 END_VAR
08 FOR i := 0 TO 27 DO
09     lo := 0;
10     FOR j := 1 TO 13 DO
11         IF features[i] >= W_DB.g0[j] THEN lo := j; END_IF;
12     END_FOR;
13     t_val := (features[i]-W_DB.g0[lo]) /
14             (W_DB.g0[lo+1]-W_DB.g0[lo]+1.0E-10);
15     FOR o := 0 TO 15 DO
16         v3[o*28+i] := W_DB.t1[o*420+i*15+lo]*(1.0-t_val)
17                   + W_DB.t1[o*420+i*15+lo+1]*t_val;
18     END_FOR;
19 END_FOR;
20 END_FUNCTION_BLOCK
```

Line numbers, indentation and syntax highlighting are rendered as a figure for print review.